

Safety Stock for Machine Error: Standby Labour, Screening Depth and Correlated Risk in LLM Services

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Abstract

Firms that deploy large language models (LLMs) create a new stochastic cost: errors that humans must absorb. We study the firm's choice of a standby human workforce and a depth of automated screening, which substitute for each other, to minimise expected cost. Standby capacity is a newsvendor quantile of the daily count of human-handled incidents, and a 95% service level is optimal only if an unhandled incident costs twenty times a capacity unit. Over-dispersion raises the safety stock beyond what the dispersion index implies, stacking screens under independence overstates diversification, and, because screening lowers the prevalence of errors that reviewers see, it erodes their vigilance, so handled-incident counts overstate success. We evaluate the model on measured outputs of three small open models through three guardrail layers, with simulated arrivals and public traces. A task-specific verifier cut required capacity by 78 to 95% but rejected 15 to 71% of correct answers, and a routed judge did not demonstrably pay for itself. Ignoring day-to-day variation raised expected cost by 9 to 65%. Arrivals are simulated, so the numbers illustrate the method.

Keywords: LLM operations, hallucination, safety stock, newsvendor, factor substitution, correlated risk, over-dispersion

1. Introduction

Firms that deploy large language models (LLMs) automate a task, answering questions, and in doing so create a new kind of random event: the answer that is wrong, or unsupported by the evidence, and that someone has to find and repair. A support assistant that invents a refund policy, or a drafting tool that cites a court decision that does not exist, turns the firm's own production process into a source of incidents. The firm then faces a problem that economists will recognise from inventory theory and from the theory of production under uncertainty. It must choose, at least expected cost, an automated screening technology, which is capital-like, scalable and imperfect, and a standby human workforce, which is labour-like, flexible and expensive, to absorb whatever the screening misses. How large should the standby workforce be, how much screening is worth buying, and how do the two substitute for one another?

Two rules of thumb are common, and both are, in economic terms, mistakes. The first is certainty-equivalent staffing: provision for the mean incident load, as if uncertainty about the load had no cost. Even when incident counts are exactly Poisson, this leaves the team short on between a quarter and a half of days once the rate reaches one event per day (Section 3.1), and in our replays it was exceeded on 28 to 49% of days. The second is the diversification fallacy: stacking guardrail layers and crediting each with its stand-alone effectiveness, so that three layers that each catch 63% of hallucinations are assumed to leave about 5%. That arithmetic holds only if the layers fail independently. Where they tend to fail on the same inputs, as correlated assets do in a portfolio, the benefit of redundancy is overstated and the residual risk is understated.

We ask three questions. First, what is the cost-minimising size of the standby workforce, and what does a given service-level target imply about the cost of an unhandled incident? Second, how does the optimal amount of automated screening depend on the relative price of human labour and compute, and how much of its apparent benefit survives once the layers' dependence and their costs in latency, false blocks and failures are taken into account? Third, do the stochastic assumptions on which the answers rest, Poisson arrivals with at most modest over-dispersion, hold up on public data from comparable real systems?

We answer them with a formal model and an empirical evaluation. The model (Section 3) treats the daily count of human-handled incidents as a random variable, sets standby capacity as a newsvendor-type quantile, and shows how over-dispersion, thinning by screening and dependence between layers shift that quantile and the cost attached to it. The evaluation (Sections 4 and 5) uses the measured outputs of three small open models that answered the same 300 SQuAD 2.0 questions through three executed guardrail layers. Arrival times are simulated, with day-to-day variability informed by public traces of an LLM service, of LLM providers' incidents, of company ticket systems and of court decisions involving fabricated citations. None of these traces, with the partial exception of the court-case database, records hallucinations, and we know of no public operator log that does.

Three findings stand out. One task-specific verifier delivered most of the benefit of the cascade: it reduced required capacity by 78 to 95%, but it also blocked 15 to 71% of correct answers, so a large part of the saving was a transfer of cost from standby labour to false rejections. A routed LLM judge did not demonstrably pay for itself under a two-second fail-open timeout. The share of hallucinations that passed two layers was 1.3 to 1.6 times the independence prediction when both layers saw every answer and 1.9 to 2.3 times when routing hid some answers from the second layer, with wide intervals. And mild day-to-day variation in demand, a coefficient of variation of 0.3 to 0.44, was enough for a Poisson-sized plan to be exceeded on 23 to 32% of days without guardrails and on 7 to 16% of days with the best cascade, against a target of 5%. In cost terms, ignoring that variation raises expected daily cost by 9.4% to 65.3% across the cases we examine (Table E1).

The contribution is mainly one of application and measurement, and we try to be clear about which results are known. The newsvendor solution and the negative-binomial treatment of over-dispersed demand are classical (Arrow et al. 1951; Porteus 2002; Cameron and Trivedi 1998), as is the Poisson quantile rule for staffing (Whitt 2007). We apply them to a new input, standby labour for machine errors, and add four things. We show how thinning by screening interacts with over-dispersion (Proposition 3). We recast the multiplicative guardrail formula as

the independence special case of dependence bounds, and measure the dependence for hallucination-grounding layers (Proposition 4). We show that, with integer capacity and a floor on standing staff, cost is generally not unimodal in the depth of screening, with comparative statics in the ratio of labour to compute prices (Proposition 5). And we replace the single-Poisson description of "all failures" by a typed model in which each class of failure follows the process that matches its mechanism (Proposition 6). Proposition 7 translates the service-level target into the shortfall cost that would rationalise it and quantifies the cost of misspecification. Proposition 8 relaxes the assumption that every surviving error is detected: it treats reviewers' attention as a moral-hazard problem in which the screening that removes errors also erodes vigilance, and it shows when gating restores it.

The paper follows the design-science tradition in IS (Hevner et al. 2004; Gregor and Hevner 2013): the artifact is a planning method, grounded in a formal model and evaluated on measured system outputs. In the terms of Gregor and Hevner (2013) the contribution is an exaptation, known solutions carried into a new problem domain. The evaluation has clear limits (Section 6.3): arrivals are simulated, the models are small and the task is extractive question answering. The numbers illustrate the method; the method and its diagnostics are what we expect to transfer. Section 2 reviews related work, Section 3 develops the model, Sections 4 and 5 describe and report the evaluation, Section 6 discusses implications and limitations, and Section 7 concludes.

2. Background and Related Work

2.1 Safety stock and staffing under random demand

Choosing capacity against random demand is the oldest problem in stochastic inventory theory. Arrow et al. (1951) derived the optimal stock as a quantile of the demand distribution, with the quantile set by the ratio of shortage and holding costs, and Porteus (2002) gives the modern treatment. In service operations the same logic sizes staffing from the quantile of a Poisson count (Whitt 2007), with piecewise-stationary adjustments for time-varying demand

(Green et al. 2007). Empirical studies of call centres find that arrivals are Poisson only conditional on a random, day-specific rate, so that daily counts are over-dispersed (Brown et al. 2005), and count-data econometrics supplies the negative-binomial remedy (Cameron and Trivedi 1998). We carry these tools over to a stream whose arrivals are created by the firm's own technology, and we test whether the usual caveats apply.

2.2 Managing AI and human oversight

Berente et al. (2021) argue that AI creates managerial problems that earlier information technologies did not, because of its autonomy, its capacity to learn and its inscrutability. A related stream examines how organisations divide work between people and AI, and describes a tension between automating tasks and augmenting human judgement (Raisch and Krakowski 2021). Human review of AI output is central to both. Yet this work says little about how large the human reserve must be, which is a question of quantities that depend on how errors arrive. Human-factors research adds that reliance on an automated aid erodes the attention people give to the cases it passes (Parasuraman and Manzey 2010), and economics treats unobservable effort as a moral-hazard problem whose solution is an incentive that depends on what can be observed (Holmström 1979). We bring the two together in Section 3.8.

2.3 Hallucination and its mitigation

Hallucination, the generation of content that is wrong or unsupported by the source, is documented as a pervasive property of neural text generation (Ji et al. 2023). Mitigation work includes retrieval grounding (Lewis et al. 2020), consistency checks based on repeated sampling (Manakul et al. 2023), safeguard classifiers (Inan et al. 2023) and verification by an LLM acting as judge (Zheng et al. 2023). This literature evaluates detectors by accuracy and does not ask what a detector's accuracy is worth in units of human capacity and money, which is the translation we attempt.

2.4 Queueing and counting-process models of AI work

Recent papers model human review of AI output, including a Poisson escalation model with Erlang-C staffing (Besanson 2026), an M/G/1 model of reviewing AI drafts (Bartolucci and Vivo 2026) and a fluid-limit analysis of routing between an LLM judge and human review (Lin and Zhang 2026). They use mean-based stability or waiting-time criteria and assume Poisson arrivals. A public-health-style analysis of AI incident databases finds strong over-dispersion in counts of vehicle and deepfake incidents (Abraham et al. 2026), and a Markov reliability model of LLM agents reduces to the Goel-Okumoto software-reliability model when failures are rare (Tran-Truong and Le 2026). We size capacity at a quantile, price it, and test the arrival assumptions.

2.5 Correlated risk in layered defences

The multiplicative formula for the residual rate of independent safeguards is standard in reliability engineering, and it over-credits redundancy when failures are positively associated (Esary et al. 1967). For LLM defence stacks, Alotaibi et al. (2026) measure positive failure correlation between seven defence layers (ϕ between 0.30 and 0.75 on JailbreakBench) and call the product a lower bound that no real stack attains, and Bhardwaj et al. (2026) give a Fréchet-Hoeffding sandwich and find high co-failure among same-model components. Our results are additional evidence in a different setting, hallucination-grounding layers, and we carry the correction through to the capacity and cost decision. The literature on LLM guardrails moved quickly during 2026, our search was not exhaustive, and similar work may exist.

2.6 Practice: gating and monitoring

Industry practice already combines automated screening with human review. Researchers at J.P. Morgan propose HalluciBot, a model that estimates a query's propensity to hallucinate before any generation, by perturbing the query, sampling outputs from several agents and running a Monte Carlo simulation to obtain an empirical hallucination rate that supervises an encoder classifier (Watson et al. 2024). Patent-analytics summaries report two filings by the

same assignee on pre-generation hallucination probability, which describe using a risk threshold to decide whether a query proceeds to the model or goes to a human expert (PatSnap n.d.); we could not obtain the full texts of those filings and cite them only as reported. A separate J.P. Morgan application describes monitoring model outputs, such as fraud scores, approval decisions and transaction counts, against the same window in earlier days or weeks and alerting an operator when they deviate (US 2024/0330824 A1). We cite these as illustrations of a gate-and-review architecture and of baseline monitoring. None of the documents we read describes hallucinations as a Poisson process or proposes to stop reducing them. They estimate risk, route risky cases to people and watch for drift, which is the structure that Sections 3.7 and 3.8 formalise.

3. Model

3.1 Environment and the firm's problem

A firm answers queries that arrive at a time-varying rate $\Lambda(t)$ with a daily cycle. Each query independently yields a hallucination with probability p_0 , and a cascade of screening layers may block some of these. Three assumptions are strong, and we revisit them in Section 6.3. First, queries are independent given the arrival intensity. Second, every hallucination that passes the whole cascade is detected and becomes exactly one human-handled event that uses one unit of standby capacity. We call such an event a *critical hallucination* and write $q \leq p_0$ for the probability that a query produces one. Third, one unit of capacity clears one critical event per day. The second assumption puts the information problem aside: the firm observes every error that gets through, so Sections 3.1 to 3.7 concern the cost of absorbing errors and not the cost of finding them. Section 3.8 relaxes it by treating the reviewers' attention as a moral-hazard problem.

The firm pays C_h per day for each unit of standby capacity, g_i per day for the compute that runs screening layer i , a false-block cost b for each correct answer a layer wrongly rejects, and a shortfall cost C_u for each critical event that finds no capacity available on the day. Its

problem is to choose capacity k and the depth N of screening to minimise expected daily cost. We first treat the service level, the probability α that capacity suffices, as a policy parameter, as practitioners do, and derive in Section 3.7 the shortfall cost that rationalises it.

Under these assumptions the thinning theorem (Kingman 1993) implies that critical hallucinations form a non-homogeneous Poisson process (NHPP) with intensity $q\Lambda(t)$. Let X_d be the number of events on day d , with mean $\lambda = q\int\Lambda(t)dt$ over the day. Then X_d is $\text{Poisson}(\lambda)$ whatever the within-day shape of Λ , which matters only for hourly staffing. If only a fraction δ of surviving hallucinations is ever noticed and triaged, the event rate is $\delta\lambda$ and, by Proposition 3, the dispersion index becomes $1 + \delta\lambda c^2$. Our capacities are therefore upper bounds that hold for $\delta = 1$. Under the NHPP the time-rescaled gaps $\Lambda(t_i) - \Lambda(t_{i-1})$ are independent $\text{Exp}(1)$ variables (Ogata 1988; Daley and Vere-Jones 2003), which yields a goodness-of-fit test used in Section 5. If the daily rate is itself random, $\Lambda_d = \lambda M$ with $E[M] = 1$ and $\text{Var}(M) = c^2$, the count is mixed Poisson with $\text{Var}(X_d) = \lambda\phi$, where $\phi = 1 + \lambda c^2$ is the dispersion index; this is the Cox, or doubly stochastic, alternative to the Poisson null (Cox 1955; Cox and Isham 1980).

For a service-level target α the capacity requirement is the generalised inverse of the count distribution:

$$k^*(\alpha, \lambda) = \min \{ k \in \mathbb{N}_0 : P(X_d \leq k) \geq \alpha \} \quad (1)$$

Proposition 1 (Poisson capacity). Let $X \sim \text{Poisson}(\lambda)$. (a) $k^*(\alpha, \lambda)$ is non-decreasing in λ . (b) $k^*(\alpha, \lambda) = 0$ if and only if $\lambda \leq -\ln \alpha$; for $\alpha = 0.95$ this threshold is 0.0513 events per day. (c) Provisioning at the mean, $k = \lfloor \lambda \rfloor$, leaves the system short with probability $P(X > \lfloor \lambda \rfloor)$, which is at least 0.26 for every $\lambda \geq 1$, about 0.35 at $\lambda = 3$, and tends to 0.5 as λ grows. (d) $k^*(\alpha, \lambda) = \lambda + z_\alpha \sqrt{\lambda} + O(1)$ as $\lambda \rightarrow \infty$.

Part (c) is the formal version of certainty-equivalent staffing: a team sized at the mean load is short on between a quarter and a half of days even when the Poisson assumption holds exactly. The proofs of all propositions are in Appendix A.

3.2 Demand uncertainty beyond Poisson

Proposition 2 (over-dispersion raises the safety stock). Let $X_d | M \sim \text{Poisson}(\lambda M)$ with $E[M] = 1$ and $\text{Var}(M) = c^2 > 0$. Then (a) $\text{Var}(X_d) = \lambda\phi$ with $\phi = 1 + \lambda c^2$. (b) X_d dominates $\text{Poisson}(\lambda)$ in the convex order, so $E[(X_d - k)^+] \geq E[(Y - k)^+]$ for every k , where $Y \sim \text{Poisson}(\lambda)$; in words, the expected shortfall at every capacity is higher than a Poisson model says. (c) If ϕ stays bounded as λ grows, $k^*_{\text{mixed}}(\alpha) = \lambda + z_{\alpha}\sqrt{\lambda\phi} + O(1)$. If instead the multiplier has a fixed coefficient of variation c , then ϕ grows with λ and $k^*_{\text{mixed}}(\alpha) = \lambda \cdot F_M^{-1}(\alpha) + O(\sqrt{\lambda})$, where F_M is the distribution of M . In that regime the normal approximation understates the exact quantile by an amount that grows linearly in λ , and $\sqrt{\phi}$ is a lower guide for the inflation of the safety stock $k^* - \lambda$, not an exact factor.

For a Gamma multiplier at $\lambda = 120.7$ and $c = 0.3$, the exact ratio of the safety stock to its Poisson value is 3.7 against $\sqrt{\phi} = 3.4$; at $c = 0.44$ it is 5.5 against 4.9. Planning should use the exact negative-binomial quantile (Table 1). At $\lambda = 10$ events per day and $c = 0.3$ the Poisson plan $k^* = 15$ is exceeded on 11.0% of days instead of 4.9%, and the correct requirement is 18; at $\lambda = 25$ and $c = 0.3$ the Poisson plan of 33 is exceeded on 16.7% of days and 41 is required.

Table 1. Capacity under over-dispersed arrivals (Gamma-mixed Poisson, $\alpha = 0.95$)

λ (events/day)	c (CV of daily rate)	ϕ	k^* Poisson	k^* mixed	P(exceed) Poisson plan
10	0	1.00	15	15	4.9%
10	0.1	1.10	15	16	5.7%
10	0.2	1.40	15	17	8.0%
10	0.3	1.90	15	18	11.0%
10	0.5	3.50	15	21	16.4%
25	0	1.00	33	33	5.0%
25	0.1	1.25	33	35	7.0%
25	0.2	2.00	33	37	11.9%
25	0.3	3.25	33	41	16.7%
25	0.5	7.25	33	50	23.0%

Exact negative-binomial quantiles, computed by `paper_tables.py`.

3.3 Screening as thinning

Requests may arrive in clusters even when incidents are rare. Because critical hallucinations are an independent thinning of requests, the dispersion of the incident stream follows from that of the request stream.

Proposition 3 (thinning reduces dispersion). Let N be the request count in a window with mean μ and Fano factor $F = \text{Var}(N)/\mu$, and let each request independently become a critical event with a constant probability p that does not depend on the request stream. Then the event count N_p has mean $p\mu$ and Fano factor $1 + p(F - 1)$. Equivalently, for a mixed-Poisson day with a rate multiplier of coefficient of variation c , $\phi = 1 + \lambda c^2$ with $\lambda = p\mu$, and independent thinning with constant p leaves M , and hence c , unchanged.

Screening therefore does two things at once: it lowers the mean number of events and pulls the stream towards Poisson. The coefficient of variation c is invariant to thinning while ϕ is not, so c is the quantity that carries over between streams with different event rates, provided p is constant and independent of M , which we assume but do not test. The practical reading is that the cost of demand uncertainty is largest exactly where capacity is largest, with weak screening, and shrinks after strong filtering.

3.4 Correlated risk and the diversification fallacy

A cascade of N layers with stand-alone clearance rates c_i , the share of critical hallucinations that layer i blocks, is usually assumed to give a residual rate $\lambda_N = \lambda_0 \Pi(1 - c_i)$. This presumes independent layer decisions. In practice the same ambiguous or adversarial outputs tend to fool all the layers, so redundancy buys less risk reduction than the formula promises, as in a portfolio of positively correlated assets.

Proposition 4 (dependence bounds). For any joint behaviour of the layers, $\lambda_0(1 - \sum c_i)^+ \leq \lambda_N \leq \lambda_0(1 - \max_i c_i)$. If the layers' catch indicators, conditional on a hallucination, are positively associated (Esary et al. 1967), then additionally $\lambda_N \geq \lambda_0 \Pi(1 - c_i)$. Consequently the multiplicative formula understates residual risk and, through Proposition 1(a), understates required capacity.

Table 2 shows how wide the identification problem is. With $c_i = 0.63$ and $\lambda_0 = 10$, three layers could imply anything from 0 to 7 units of capacity depending on the unknown dependence, so the dependence has to be measured (Section 5.2) and not assumed. Table E2 prices the error in Section 3.7.

Table 2. Residual rate and capacity bounds ($\lambda_0 = 10$, $c_i = 0.63$, $\alpha = 0.95$)

N	λ_N lower	λ_N independent	λ_N upper	k^* lower	k^* indep.	k^* upper
1	3.70	3.70	3.70	7	7	7
2	0.00	1.37	3.70	0	4	7
3	0.00	0.51	3.70	0	2	7
4	0.00	0.19	3.70	0	1	7

3.5 Substituting compute for labour: the optimal depth of screening

Let $k_{\min} \geq 0$ be a minimum standing capacity, since an on-call engineer is not dismissed because incidents are rare. The firm's cost under a service-level constraint is

$$f(N) = C_h \cdot \max\{k_{\min}, k^*(\alpha, \lambda_N)\} + \sum_{i \leq N} g_i \quad (2)$$

subject to a latency constraint $\sum_{i \leq N} L_i \leq \text{SLA}$, where L_i is the added latency of layer i . Screening is a capital-like input that lowers the quantity of labour required, so the problem is one of factor substitution.

Proposition 5 (optimal depth and factor substitution). (a) In general f is not unimodal in N : because k^* is integer-valued with plateaus, f can rise and then fall again. (b) With $k_{\min} \geq 1$, let N_0 be the first N with $k^*(\alpha, \lambda_N) \leq k_{\min}$; every $N > N_0$ is strictly dominated, so $N \leq N_0$. (c) Let $\theta = C_h/g$ with equal per-layer compute cost g . The minimiser $N(\theta)$ is non-decreasing in θ : the dearer labour is relative to compute, the deeper the optimal cascade. (d) The latency constraint restricts N to a feasible prefix of the cascade.

Table 3 uses an illustrative parameter set ($\lambda_0 = 10$ events per day, a clearance of 0.63 per layer under independence, \$150 per capacity unit, \$25 per layer). With $k_{\min} = 0$ the cost falls from $N = 3$ to $N = 4$, rises at $N = 5$ and falls again at $N = 6$, where k^* reaches zero ($\lambda_6 = 0.026 < -\ln 0.95$), so the minimum is at $N = 6$. With a floor of one on-call engineer ($k_{\min} = 1$) the minimum is at $N = 4$. The non-monotone cost schedule is the signature of indivisibility: capacity moves in whole units, so marginal analysis finds only local optima. Table E3 gives the comparative statics: as the price of labour rises from the price of compute to many times it, the optimal depth rises from three layers to four or six and then stays there, because the floor and the

integer plateau bind. A fixed prescription such as "three layers" is therefore not implied by the cost model.

Table 3. Daily cost by guardrail depth ($\lambda_0 = 10$, $c = 0.63$ independent, $C_h = \$150/\text{unit}$, $g = \$25/\text{layer}$)

N	λ_N	k^*	Labour	Compute	Total ($k_{\min} = 0$)	Total ($k_{\min} = 1$)
0	10.000	15	\$2,250	\$0	\$2,250	\$2,250
1	3.700	7	\$1,050	\$25	\$1,075	\$1,075
2	1.369	4	\$600	\$50	\$650	\$650
3	0.507	2	\$300	\$75	\$375	\$375
4	0.187	1	\$150	\$100	\$250	\$250
5	0.069	1	\$150	\$125	\$275	\$275
6	0.026	0	\$0	\$150	\$150	\$300
7	0.009	0	\$0	\$175	\$175	\$325

Minimum: $N = 6$ under $k_{\min} = 0$; $N = 4$ under $k_{\min} = 1$. The cost parameters are illustrative assumptions.

Table E3. Optimal depth of screening against the labour-to-compute price ratio $\theta = C_h/g$ ($\lambda_0 = 10$, $c_i = 0.63$, $\alpha = 0.95$, $g = \$25$)

C_h (\$/unit/day)	θ	$N^* (k_{\min} = 0)$	$N^* (k_{\min} = 1)$
25	1	3	3
50	2	4	4
100	4	6	4
150	6	6	4
300	12	6	4
600	24	6	4
1200	48	6	4

Computed by `econ_tables.py`; the cascade is evaluated under independence.

3.6 Pooling heterogeneous risks: typed failure classes

An LLM service produces several kinds of failure, and they violate the Poisson assumptions in different ways. Treating all failures as one Poisson stream is therefore a modelling error and not a harmless simplification: the variances of independent classes add, but each class has its own dispersion, and the pooled stream is Poisson only if every component is. We model each class with the process that matches its mechanism and combine them only at the level of the human-handled event, an incident or a ticket (Table 4).

Table 4. Failure classes, mechanisms and models

Class	Mechanism	Poisson assumption violated	Model used	Evidence
Hallucination (per answered query)	Independent errors of the model given the query, on a clustered, daily-varying request stream	Constant rate; independence of requests	Cox / negative-binomial thinning of the request stream (Props. 2, 3)	Sections 5.3, 5.5
Provider-declared incidents	Outages, possibly followed by further incidents within hours; weekday and hour seasonality and a drifting level	Orderliness and independence (self-excitation)	Seasonal, level-adjusted Hawkes process with exponential kernel; NHPP as null	Section 5.7, App. C
Serving errors and timeouts	Common-cause episodes (outages, rate limiting) with heavy-tailed size	Orderliness (batch arrivals), independence	Episode process (Hawkes) with heavy-tailed marks; compound Poisson for failed-request counts	Section 5.7, App. C
Human triage service	Mixture of quick and very long tickets	Exponential service	Heavy-tailed (Weibull, lognormal) service; workload headroom	Section 5.6

Proposition 6 (superposition and clustering). (a) If the daily counts of independent failure classes j have means λ_j and dispersion indices ϕ_j , the total daily count has mean $\Sigma \lambda_j$ and variance $\Sigma \lambda_j \phi_j$, so the 95% capacity is approximately $\Sigma \lambda_j + z \cdot \sqrt{\Sigma \lambda_j \phi_j}$. A single Poisson on the total uses $\phi = 1$ and understates the safety stock by the factor $\sqrt{\phi_{\text{eff}}}$, where $\phi_{\text{eff}} = \Sigma \lambda_j \phi_j / \Sigma \lambda_j$. (b) For a stationary Hawkes process (Hawkes 1971) with branching ratio $\alpha < 1$, the cluster started by a background event has a Borel size distribution with mean $1/(1 - \alpha)$, and the long-window Fano factor of counts is $1/(1 - \alpha)^2$. (c) For a compound Poisson count $N = \Sigma_{i \leq M} X_i$ with $M \sim \text{Poisson}(v)$, the Fano factor is $E[X^2]/E[X]$, which is unbounded when the mark distribution has an infinite second moment. The variance of the daily count, and with it the error of a Poisson plan, is then dominated by the largest episodes.

3.7 The implied cost of a shortfall, and the cost of misspecification

Sections 3.1 to 3.6 took the service level α as given. An economist will ask what shortfall cost would make it optimal. Suppose that each critical event that finds no capacity costs the firm C_u , for example through delay, escalation or harm to the customer, and that capacity costs C_h per unit whether or not it is used. The firm minimises $\Phi(k) = C_h k + C_u E[(X - k)^+]$.

Proposition 7 (newsvendor service level). (a) Φ is convex in k , and its smallest minimiser is the α° -quantile of X with $\alpha^\circ = 1 - C_h/C_u$, so that a service-level target α is cost-minimising exactly when $C_u = C_h/(1 - \alpha)$. (b) If $C_u \leq C_h$ the optimal

capacity is zero. (c) If the firm plans with a Poisson model when the true count is mixed Poisson, its perceived cost curve lies below the true one at every k , by Proposition 2(b), and the plan it chooses is generally too small. (d) The minimal expected cost $\Phi^\circ(C_h)$ is concave in the capacity price, and $d\Phi^\circ/dC_h = k^\circ$ wherever the minimiser is unique: the shadow price of a unit of capacity is the capacity held.

With $C_h = \$150$ per unit and per day, a 95% service level is optimal if an unhandled event costs $C_u = \$3,000$, a 90% level if it costs $\$1,500$, and a 99% level if it costs $\$15,000$. The standard target of 95% therefore embodies a specific, checkable claim about how costly an unhandled event is, twenty times the price of the unit that would have handled it. The scenario residual-harm value of $\$100$ per hallucination that reaches a user, used in Section 4.4, would rationalise no standby capacity at all if it were the shortfall cost, which shows that the two costs describe different events: harm from an error the user receives, and delay in handling an error the firm has detected.

Table E1 prices the cost of misspecification at $C_u = \$3,000$. When the true daily count has a coefficient of variation of the rate of 0.3, a firm that plans with a Poisson model holds 15 units instead of 18 at $\lambda = 10$ and bears 9.4% more expected cost than necessary; at $\lambda = 120.7$ it holds 139 instead of 189 and bears 43.6% more. Staffing at the mean is much worse, with excess costs around 102 to 121%. The loss grows with the rate and with the dispersion, in line with Propositions 2 and 3.

Table E1. Expected daily cost of planning errors ($C_h = \$150$, $C_u = \$3,000$, optimal service level 95%)

λ (events/day)	c	k optimal (true)	k Poisson plan	k mean plan	Optimal cost	Excess: Poisson plan	Excess: mean plan
10	0.3	18	15	10	\$3,070	+9.4%	+117%
10	0.44	20	15	10	\$3,558	+20.7%	+121%
25	0.3	41	33	25	\$6,983	+18.9%	+107%
25	0.44	47	33	25	\$8,335	+36.2%	+116%
120.7	0.3	189	139	120	\$31,659	+43.6%	+102%
120.7	0.44	221	139	120	\$38,669	+65.3%	+114%

True counts are negative binomial (Gamma-mixed Poisson). Costs are evaluated under the true distribution. Computed by econ_tables.py.

Table E2 prices the diversification fallacy for a cascade of N identical layers that each clear 63% of hallucinations ($\lambda_0 = 10$). If the firm plans for independent layers but the layers in fact fail together, the extreme case of the upper bound in Proposition 4, the shortfall cost is many times the planned cost. With three layers the firm holds 2 units and expects to pay \$351 per day; under the comonotone upper bound the same plan costs \$5,823, against an optimum of \$1,218 with 7 units. The upper bound is an extreme case, and the measured dependence in Section 5.2 is far weaker, but it shows why deep cascades with strong dependence are the setting in which the independence assumption is most expensive.

Table E2. Cost of assuming independent layers when they fail together ($\lambda_0 = 10$, clearance 0.63 per layer, $C_h = \$150$, $C_u = \$3,000$)

N	λ_N independent	λ_N upper bound	k planned (independence)	Cost if independence holds	Cost if layers fail together	Optimal k / cost if layers fail together
2	1.37	3.70	4	\$650	\$2,462	7 / \$1,218
3	0.51	3.70	2	\$351	\$5,823	7 / \$1,218
4	0.19	3.70	1	\$200	\$8,324	7 / \$1,218

Poisson counts in both cases. The upper bound is the comonotone case of Proposition 4. Computed by econ_tables.py.

3.8 Moral hazard in review: screening erodes vigilance

Sections 3.1 to 3.7 assumed that every error that passes the screens is detected. In practice detection is done by people whose attention the firm cannot observe, and whose incentive to attend depends on how often attention pays. Suppose a reviewer inspects M items per day, of which a share $\rho = \lambda_N/M$ are critical hallucinations. On each item the reviewer chooses attention $e \in [0, 1]$ at a private cost $\psi e^2/2$ and finds an error, if there is one, with probability e . The reviewer is rewarded r , in money or in reputation, for each error found. The firm observes neither e nor the errors that are missed, which is the standard setting of moral hazard (Holmström 1979).

Proposition 8 (vigilance under screening and gating). (a) The reviewer's optimal attention is $e = \min\{1, r\rho/\psi\}$. (b) For a fixed reward r , e is non-decreasing in the prevalence ρ . Adding screening layers lowers ρ and therefore weakly lowers detection, and in the interior region the number of errors found, $e^*\lambda_N = r\lambda_N^2/(\psi M)$, falls with the square of the residual rate. (c) Holding attention at $e = 1$ requires $r \geq \psi/\rho$, a reward that rises in inverse proportion to prevalence, while the expected payment at full attention,

$rpM = \psi M$, does not depend on screening depth. (d) Consider a gate that sends a fraction γ of items to review and contains a share s of the critical hallucinations. It raises prevalence in the reviewed set to sp/γ and, when attention is interior with and without the gate, raises the expected number of errors found if and only if $s^2 > \gamma$.

Part (b) is the vigilance problem in economic form. Screening and attention are substitutes in effect: the better the automated layers, the rarer an error is among the items a person sees, and the less a reviewer gains from looking hard. The number of events handled, which is the figure a planner observes, then falls faster than the number of errors that survive. Table E4 illustrates this with the cascade of Table 3 scaled to $\lambda_0 = 100$ errors per day among 1,000 reviewed items, a reward of \$7.50 per error found and an effort-cost scale of \$0.30 per item. Without screening, prevalence is 10% and reviewers attend fully. With three layers the residual rate is 5.1 per day, prevalence 0.51% and attention 0.13, so the team handles 0.64 events per day while 4.4 errors a day go undetected. A capacity plan that assumes full detection holds 9 units, although 2 would be used, and an operations dashboard that counts handled incidents would show a fall of 99.4% from the unscreened level, against a true fall of 94.9% in surviving errors. The metric that planners watch therefore overstates the success of screening, and the approach of maintaining the status quo while watching incident counts hides the loss of detection.

Table E4. Reviewer attention and detection by screening depth ($\lambda_0 = 100$, clearance 0.63 per layer, $M = 1,000$ items per day, reward $r = \$7.50$, $\psi = \$0.30$, harm \$100 per undetected error, $C_h = \$150$, $\alpha = 0.95$)

N	Residual rate λ_N	Prevalence ρ	Attention e^*	Events handled	Undetected errors	k planned (full detection)	k actually needed	Reward for full attention
0	100.0	10.00%	1.00	100.00	0.0	117	117	\$3.0
1	37.0	3.70%	0.92	34.22	2.8	47	44	\$8.1
2	13.7	1.37%	0.34	4.69	9.0	20	8	\$22
3	5.1	0.51%	0.13	0.64	4.4	9	2	\$59
4	1.9	0.19%	0.05	0.09	1.8	4	1	\$160
5	0.7	0.07%	0.02	0.01	0.7	2	0	\$433
6	0.3	0.03%	0.01	0.00	0.3	1	0	\$1169

Illustrative parameters; computed by econ_tables.py. Attention is $e^ = \min\{1, rp/\psi\}$; events handled are $e^*\lambda_N$; undetected errors are $(1 - e^*)\lambda_N$. The reward for full attention is ψ/p , and its expected payment is $\psi M = \$300$ per day at every depth.*

Part (c) says what it takes to restore vigilance. At three layers the reward needed for full attention is \$59 per error found instead of \$3 without screening, and at four layers it is \$160, although the daily payment stays at \$300. The firm can therefore sustain attention by paying more per error, by paying a flat amount for review, which is what the constant daily payment amounts to, or by raising prevalence in what the reviewer sees. Part (d) is the last route and corresponds to the gate-and-review architecture of Section 2.6. For a gate that sends 20% of items to review and holds 70% of the errors, $s^2/\gamma = 2.45$, and at three layers the errors found rise from 0.64 to 1.57 per day. An uninformative gate, with $s = \gamma$, lowers detection. The gate thus has a second benefit that the capacity model alone does not show: besides lowering the review volume, it concentrates the errors in the cases people see, which keeps their attention worth its cost.

4. Empirical Evaluation: Design

4.1 Data, generators and ground truth

We use the validation set of SQuAD 2.0 (Rajpurkar et al. 2018), which mixes answerable questions with questions that the supplied passage cannot answer. One random sample of 300 questions (seed 0; 148 unanswerable) was drawn once and given to all three generators, so the three runs are paired on the same questions and are not independent samples. Each model receives the passage and the question and must reply with a short answer or the token UNANSWERABLE. A response is labelled a hallucination if it is not an abstention and either the question is answerable and the answer matches no gold answer (normalised containment, or token-F1 of at least 0.6), or the question is unanswerable. Abstentions are never counted as hallucinations and bypass the guardrails. The generators are Qwen2.5-0.5B-Instruct, SmolLM2-360M-Instruct and Qwen2.5-1.5B-Instruct (Qwen Team 2024; Allal et al. 2025), run greedily on a CPU. We chose small open models because they can be run end to end with full control over latency measurement. Their hallucination rates are higher than those of frontier models, so absolute rates should not be transferred (Section 6.3).

4.2 Guardrail layers

The three layers follow the usual progression from deterministic rules, through a trained classifier, to a semantic judge, and each one is implemented and executed on every answer. Layer L1 applies regular expressions for e-mail addresses, national identifiers, card and phone numbers, checks for template-token leakage and degenerate repetition, and blocks any answer containing a number that is absent from the evidence passage. Layer L2 is an extractive question-answering verifier: a SQuAD 2.0-trained model (deepset/roberta-base-squad2) answers the question from the passage, the score is the token-F1 between its span and the generator's answer (zero if it predicts no answer), and the output is blocked when the score is below 0.5, a threshold we did not tune. Layer L3 is an LLM judge (Qwen2.5-1.5B-Instruct in every run) that sees the passage, the question and the answer and replies SUPPORTED or UNSUPPORTED (Zheng et al. 2023); it is invoked only when routed, which with L2 present means only if the L2 score is below 0.9.

The first implementation of L2 was an NLI cross-encoder (DeBERTa-v3-small; He et al. 2021) that scored whether the passage entails "the answer to Q is A". It proved uninformative, and we replaced it with the extractive verifier after seeing the results (Section 5.1). We did not use safeguard classifiers of the Llama Guard type because they target policy violations and not factual grounding. All layers use the same evidence passage, as in retrieval-grounded verification. Because every layer is run on every example, any subset or cascade can be evaluated offline from the observed flags, without an independence assumption. For each call we record the flag, the score, the wall-clock latency and any exception.

Layers fail open. A call that raises an exception or takes longer than 2 s is counted as a system failure, from which we estimate a failure rate and not assume one; its verdict is discarded, so the answer passes that layer, and the added latency is capped at 2 s. No call raised an exception, so all system failures are judge calls slower than 2 s, which were 25 to 30% of judge calls on our CPU set-up. Section 5.4 reports a sensitivity analysis in which slow verdicts are awaited and credited.

4.3 Traffic replay and estimators

Production logs of hallucinations are unavailable, so arrivals are simulated while outcomes are measured. Queries arrive according to an NHPP with a 24-slot hour-of-day intensity and a mean of 200 queries per day over 90 days. The default profile follows business hours; a second profile is the hour-of-day profile measured in the BurstGPT production trace (Section 4.5). A Cox variant multiplies each day's intensity by a Gamma factor with mean 1 and coefficient of variation c , to represent bursty demand. We use $c = 0.3$ and $c = 0.44$ as scenario values. Net of trend, and after excluding holiday and low-volume days, public ticket streams give daily-rate coefficients of variation of 0.10 to 0.38 (0.25 to 0.41 with all days; Section 5.6), so $c = 0.3$ lies within that range and $c = 0.44$ above it. The value 0.44 is the smallest coefficient of variation of any 14-day window of BurstGPT (Section 5.5), whose variability over the whole trace is much larger, so it is a floor for that trace and not a net-of-trend estimate comparable to the ticket values.

Each arriving query replays a uniformly drawn example from the measured pool, and it becomes a critical event if it is a hallucination that survives the cascade (the detection assumption of Section 3.1). For each event stream we estimate $\hat{\lambda}$ as events per day and apply four tests: an index-of-dispersion test on daily counts, a chi-square test of constant hourly intensity, a Kolmogorov-Smirnov test of raw inter-arrival times against an exponential and of time-rescaled gaps under the fitted hourly intensity (the Lilliefors-type p-value is optimistic), and a negative-binomial fit of daily counts. Uncertainty from the finite evaluation pool is quantified with a 300-draw bootstrap over examples, reported as 95% percentile intervals. Because the same pool generates all arrivals, the replay tests whether the counting process has the claimed structure given measured per-query outcomes. It does not test whether real users' queries are independent, which is the main limit on external validity.

4.4 Cascade selection and cost model

We evaluate the cascades L1, L2 and L3 and all their combinations, always in the order L1, L2, L3, including the case of no guardrail. The *best cascade* is the one that minimises $k^* + 0.02 \times (\text{mean added latency in ms}) + (\text{system failures per day})$ among cascades whose mean added latency is within the stated budget (100 ms, or 500 ms where indicated). The weights are arbitrary and were not tuned; they serve only to produce one ranking for Sections 5.3 and 5.7, where L1+L2 under the 500 ms budget is selected for all three generators.

Daily cost is $C_h \cdot \max(k_{\min}, k^*)$ plus compute, false-block and residual-harm costs. We set C_h to \$150 per capacity unit, compute to \$0.0001 per classifier call and \$0.002 per judge call, the false-block cost to \$1 per wrongly blocked answer and the residual-harm cost to \$100 per hallucination that reaches a user. These are scenario values, not estimates. The *minimum-dollar cascade* of Section 5.4 minimises this cost and can differ from the best cascade because it ignores latency and system failures. A sensitivity grid varies daily traffic (50, 200 and 800 queries), the false-block cost (\$0, \$1, \$10) and the residual-harm cost (\$0, \$100, \$1,000), giving 27 scenarios per generator. Sampling uncertainty is quantified by pool bootstrap, including paired intervals for cost differences between cascades.

4.5 Public production data

Six public sources test the counting-process assumptions on real systems. BurstGPT (Wang et al. 2025) records 10.6 million requests to regional Azure OpenAI services over 231 days. A dataset of LLM service outages and incident reports (Chu et al. 2025) provides 457 incidents declared by three providers from March 2023. Three company ticket logs describe human triage: customer tickets of an international software company (Abdellatif 2026), the incident log of a ServiceNow instance (Amaral et al. 2018) and the help-desk log of an Italian software company (Verenich 2016). Finally, a curated database of court decisions involving fabricated content (Charlotin 2026) is the only dated log of hallucination incidents we found, although it is confined to the legal domain and is subject to strong detection and reporting bias.

Appendix B gives the filters, the work-week and holiday rules, and the estimators. Two features of the design are worth stating here. Each ticket analysis is reported twice, once for all work days and once with holidays and low-volume days excluded, because isolated dips inflate dispersion that a local mean cannot absorb. And each Poisson plan is evaluated both against a two-sided local mean, which is not a feasible forecast, and causally, with a trailing mean and a coefficient of variation estimated on earlier days only.

For typed failure models we fit a seasonal Hawkes process with exponential kernel to provider incidents, $\lambda(t) = \mu_0 s(t) + \sum \alpha \beta e^{-\beta(t - t_i)}$, where $s(t)$ is a weekday-by-hour profile estimated from the data and held fixed, and compare it with a seasonal NHPP by AIC, by a parametric-bootstrap likelihood-ratio test and by the time-rescaling test. For BurstGPT failures we detect outage-like episodes, characterise their size and arrival process, and test four alternative episode definitions. Finally, we combine class-specific negative-binomial daily counts by Monte Carlo to compare the 95% capacity of a typed superposition with that of a single Poisson on the total (Section 5.7). All simulations and bootstrap draws are seeded and can be reproduced with the scripts that accompany the paper (Appendix D).

5. Findings

5.1 Hallucination rates and guardrail layer performance

On the same 300 questions the hallucination rate is 60.3% for Qwen2.5-0.5B, 65.0% for SmolLM2-360M and 47.0% for Qwen2.5-1.5B, and 75 to 90% of the hallucinations are answers to unanswerable questions. Table 5 reports the layers. The deterministic layer L1 catches only 0.7 to 5.1% of hallucinations: the task contains no personal data or template leaks and few fabricated numbers, so L1 is nearly inert, though also nearly free of latency and false blocks. The judge L3 blocks 48 to 59% of hallucinations and 17 to 24% of correct answers, and each call takes about 1.8 to 1.9 s on our CPU. Because 25 to 30% of judge calls exceed the 2 s timeout and their verdicts are discarded, these clearances are lower than they would be if slow verdicts were

credited (Section 5.4). The judge is Qwen2.5-1.5B-Instruct throughout, so one arm is judged by its own generator; we did not test for a self-judging effect.

The NLI classifier we tried first gave a negative result. Its entailment score separated correct from hallucinated answers with an AUC of 0.52 to 0.58, where 0.5 is chance, and at the default threshold it blocked 87 to 88% of hallucinations and 85 to 90% of correct answers. Its apparent capacity savings were an artefact of blocking almost everything, which is why we report false-block rates next to every capacity figure: a capacity reduction alone cannot distinguish a useful layer from one that blocks indiscriminately. We replaced it with the extractive verifier (AUC 0.84 to 0.90) and re-ran all layers on the stored answers. The replacement was made after seeing the results, the verifier is evaluated on the same 300 questions, and it is in effect a second solver of the task it verifies, trained on SQuAD, which flatters it relative to a generic classifier. All numbers below use it.

Table 5. Guardrail layers on the same 300 SQuAD 2.0 questions for each generator (CPU inference; latency re-measured without concurrent jobs)

Generator	p_0 (hallucination rate)	c_1 (L1)	c_2 (L2 QA)	c_3 (L3 judge)	L2 false-block (good answers)	L3 false-block	L2 mean ms	L3 mean ms (capped at 2 s)
Qwen2.5-0.5B	60.3%	2.8%	89.5%	58.0%	15.1%	24.4%	98	1759
SmolLM2-360M	65.0%	5.1%	96.9%	48.2%	70.9%	16.5%	97	1792
Qwen2.5-1.5B	47.0%	0.7%	81.6%	58.9%	15.4%	22.8%	95	1728

c_i is the stand-alone share of hallucinated answers the layer blocks. False-block is the share of correct, non-abstained answers blocked (denominators 119, 103 and 136). The L2 threshold ($F1 < 0.5$) was not tuned; SmolLM2 gives longer answers, which lowers its token-F1 agreement and inflates its L2 false-block rate.

5.2 Dependence between layers and the cost of routing

Among hallucinated answers, the catch decisions of L2 and L3 are weakly and positively associated ($\phi = 0.15, 0.11, 0.12$ for the three generators; 95% bootstrap intervals [0.00, 0.29], [-0.02, 0.21], [-0.06, 0.28]; Fisher exact $p = 0.054, 0.213, 0.186$), which is the direction Proposition 4 predicts. The intervals include zero for two of the three generators and the runs

share their questions, so the evidence is suggestive and not conclusive. L1 catches only 1 to 10 hallucinations per generator, so its pairwise statistics ($\varphi \leq 0.09$) say nothing useful.

The excess of the residual share over the independence formula is correspondingly modest (Figure 1, left). For the L2+L3 cascade, the share of hallucinations that pass both layers is 1.6 to 7.6% under the independence formula, computed from the same sample's marginals, and 2.6 to 9.9% when both layers see every answer. That is 1.5, 1.6, 1.3 times higher (12, 5, 14 hallucinations pass; pool-bootstrap intervals [1.0, 2.0], [0.9, 2.1], [0.8, 1.7]), and the interval includes 1 for 2 of the three generators. The cost of dynamic routing, in which only low-agreement answers go to the judge, is clearer. The ratios rise to 2.1, 1.9, 2.3 ([1.7, 2.6], [1.7, 2.2], [1.9, 3.0]), and the number of passing answers rises to 17, 6, 25. Routing cuts judge calls by 89 to 92%, but the judge no longer sees the answers the verifier considers consistent, and those contain a substantial share of the hallucinations the judge would have caught. The absolute numbers of passing answers are small, so all of these estimates are imprecise.

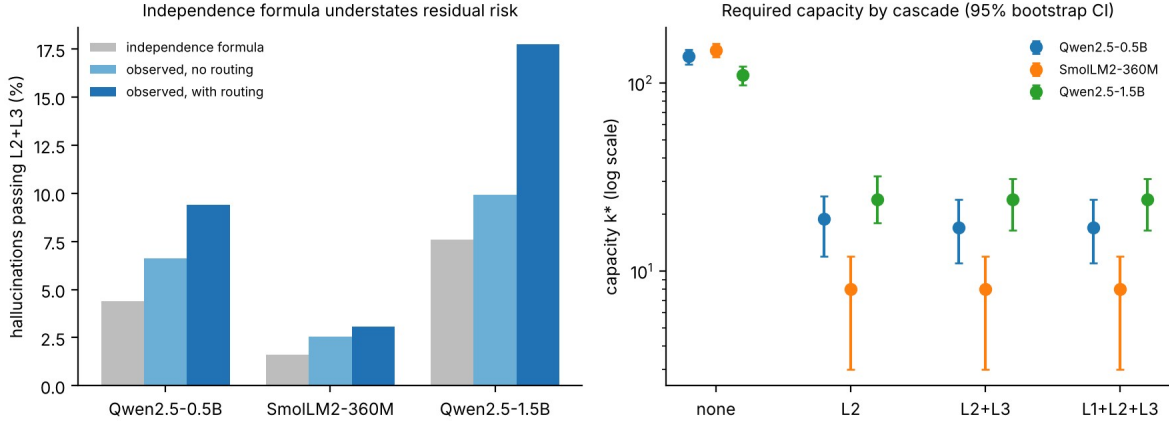


Figure 1. Left: share of hallucinations passing the L2+L3 cascade under independence, observed without routing, and observed with routing. Right: capacity k^* by cascade (95% bootstrap CI over the evaluated pool), log scale.

5.3 Burstiness: Poisson-sized plans under Cox traffic

With non-homogeneous Poisson arrivals and no burstiness the replay reproduces the Poisson structure, as it must by construction. For the unprotected stream the daily-count dispersion test does not reject ($p = 0.53$ to 0.91), raw inter-arrival times are strongly non-

exponential because the intensity varies by the hour, and the time-rescaled gaps are consistent with $\text{Exp}(1)$ ($p = 0.50$ to 0.85). This checks the replay and the estimators and shows that the rescaling machinery behaves as theory predicts. It is not evidence that real traffic is Poisson, because the arrivals are simulated.

When the daily intensity is multiplied by a Gamma factor with a coefficient of variation of only 0.3, the dispersion test rejects overwhelmingly ($p < 10^{-90}$ for the unprotected stream) and the negative-binomial shape parameter r lies between 12.7 and 13.4. The Poisson capacity $k^* = 136, 146, 108$ for the three generators without guardrails is then exceeded on 23 to 28% of days instead of the targeted 5%. The correct negative-binomial requirement is 178, 194, 141, that is, 31 to 33% more (Figure 2). Under the best cascade ($L1+L2$, λ of about 4 to 17 per day) the effect is smaller in absolute capacity, but the Poisson plan is still exceeded on 10 to 11% of days (Table 6). Because the burstiness is injected, this replay confirms Proposition 2 and Table 1 on measured outcomes and is not an independent empirical finding. Provisioning at the mean was exceeded on 28 to 49% of days across all configurations we replayed, against the 5% target of the quantile plan.

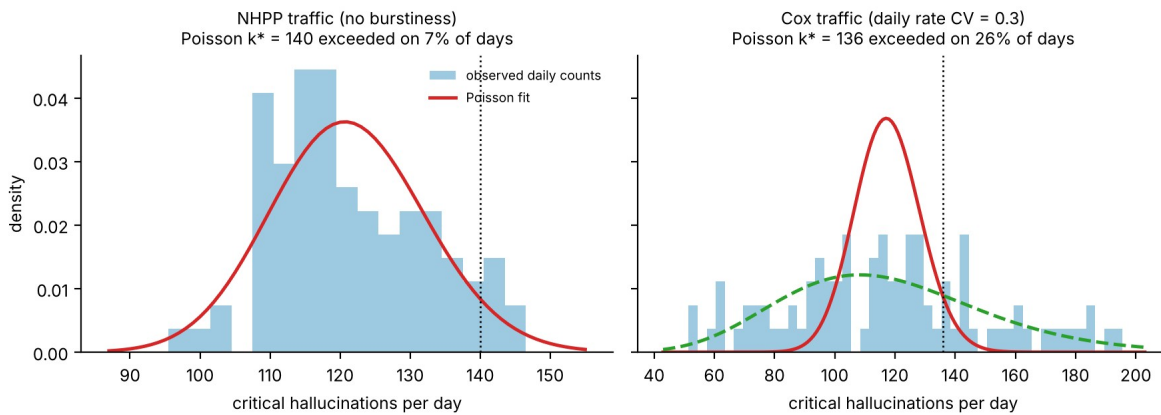


Figure 2. Daily critical hallucinations without guardrails (Qwen2.5-0.5B, 90 days, 200 queries/day). Left: steady NHPP traffic. Right: bursty traffic. Dotted line: Poisson k^* ($\alpha = 0.95$).

To place the scenario values in context, the smallest coefficient of variation of any 14-day window of BurstGPT is 0.44, a floor for a trace whose median window has 0.91 (Section

5.5), while the daily-rate coefficient of variation net of trend in public ticket streams is 0.10 to 0.38 (Section 5.6; 0.25 to 0.41 before exclusions). Table 6 gives the exceedance rates under the two scenario values, the second combined with the BurstGPT hour-of-day profile. The gap shrinks with guardrails but does not vanish: for the best cascade the Poisson plan is exceeded on 7 to 16% of days at $c = 0.44$, against 30 to 32% without guardrails. This is what Propositions 2 and 3 lead one to expect. For a given c , $\phi = 1 + \lambda c^2$ is smaller when the event rate is smaller, so over-dispersion inflates the safety stock by fewer units (about 5, 1, 8 extra), although the exceedance probability remains between 1.3 and 3 times the target.

Table 6. Share of days on which the Poisson-sized capacity k^* is exceeded (target 5%; 90 simulated days)

Generator	Cascade	Steady NHPP ($c = 0$)	$c = 0.3$	$c = 0.44$ + BurstGPT profile	k^* Poisson $\rightarrow$ k^* negative-binomial ($c = 0.44$)
Qwen2.5-0.5B	none	6.7%	25.6%	32.2%	135 $\rightarrow$ 209
Qwen2.5-0.5B	L1+L2	3.3%	10.0%	14.4%	18 $\rightarrow$ 23
SmolLM2-360M	none	5.6%	27.8%	32.2%	144 $\rightarrow$ 225
SmolLM2-360M	L1+L2	5.6%	11.1%	6.7%	8 $\rightarrow$ 9
Qwen2.5-1.5B	none	8.9%	23.3%	30.0%	106 $\rightarrow$ 162
Qwen2.5-1.5B	L1+L2	3.3%	10.0%	15.6%	24 $\rightarrow$ 32

Observed over 90 simulated days, so each share has a standard error of about 2 to 5 percentage points. L1+L2 is the best cascade by the objective of Section 4.4 within the 500 ms budget; it is not the minimum-dollar cascade of Section 5.4.

5.4 Capacity, latency and cost by cascade

Table 7 gives, for each generator, the residual rate, the capacity, the added latency and the volume of false blocks for the main cascades at 200 queries per day, with bootstrap intervals over the evaluated pool. At a pool of 300 questions the intervals are wide, so differences of a few units of k^* between cascades are not significant; the large differences are.

Table 7. Cascade performance at 200 queries/day, $\alpha = 0.95$, SLA 500 ms mean added latency (95% bootstrap CI in brackets)

Generator	Cascade	λ per day	k^*	Mean added latency (ms)	False blocks / day	System failures / day
Qwen2.5-0.5B	none	120.7 [109, 132]	139 [126, 151]	0	0.0	0.0
Qwen2.5-0.5B	L1+L2	12.7 [7, 18]	19 [12, 25]	97	12.7	0.0
Qwen2.5-0.5B	L2	12.7 [7, 18]	19 [12, 25]	98	12.0	0.0
Qwen2.5-0.5B	L3	50.7 [39, 60]	63 [49, 73]	1759 ✗	19.3	60.0

Generator	Cascade	λ per day	k*	Mean added latency (ms)	False blocks / day	System failures / day
Qwen2.5-0.5B	L2+L3	11.3 [7, 17]	17 [11, 24]	287	16.7	6.0
SmolLM2-360M	none	130.0 [120, 142]	149 [138, 162]	0	0.0	0.0
SmolLM2-360M	L1+L2	4.0 [1, 7]	8 [3, 12]	93	48.7	0.0
SmolLM2-360M	L2	4.0 [1, 7]	8 [3, 12]	97	48.7	0.0
SmolLM2-360M	L3	67.3 [58, 78]	81 [71, 94]	1780 ✗	11.3	50.0
SmolLM2-360M	L2+L3	4.0 [1, 7]	8 [3, 12]	249	50.7	6.7
Qwen2.5-1.5B	none	94.0 [82, 106]	110 [97, 123]	0	0.0	0.0
Qwen2.5-1.5B	L1+L2	17.3 [12, 24]	24 [18, 32]	87	14.0	0.0
Qwen2.5-1.5B	L2	17.3 [12, 24]	24 [18, 32]	88	14.0	0.0
Qwen2.5-1.5B	L3	38.7 [29, 48]	49 [39, 60]	1596 ✗	20.7	49.3
Qwen2.5-1.5B	L2+L3	16.7 [11, 23]	24 [16, 31]	279	19.3	6.7

L2+L3 is the routed cascade (the judge is called only if the L2 score is below 0.9); L3 alone runs the judge on every answer. ✗ marks a violation of the 500 ms budget. Mean added latency is per query, including abstentions, which bypass the guardrails, so it differs from the per-layer means of Table 5; each judge call is capped at the 2 s timeout, after which its verdict is discarded and it counts as a system failure. L1+L2 and L1+L2+L3 equal L2 and L2+L3 up to small differences because L1 is nearly inert on this task.

A single task-specific verifier (L2) reduces required capacity from 139, 149, 110 to 19, 8, 24 units per day, a reduction of 78 to 95%. Adding the routed judge changes it to 17, 8, 24, a difference of 2, 0, 0 units, while it roughly triples added latency and adds about six system failures per day through timeouts. Running the judge on every answer is infeasible under any latency budget we consider (about 1.6 to 1.8 s per query) and produces 50 to 60 system failures per day. Under a 100 ms budget only L1 and L2 are feasible (L2 takes 90 to 100 ms here); under 500 ms the routed judge fits on average but not at the 95th percentile (1.8 to 1.9 s). Latencies would be much lower on GPUs or hosted inference.

The best cascade by the objective of Section 4.4 is L1+L2 for all three generators. At the scenario costs the minimum-dollar cascade is L1+L2+L3 for Qwen2.5-0.5B, L1+L2 for SmolLM2-360M, L1+L2+L3 for Qwen2.5-1.5B, and the daily cost falls from \$32,917, \$35,350, \$25,900 without guardrails to \$3,700, \$1,649, \$5,286 at the minimum. The paired bootstrap interval of the daily saving from adding the routed judge to L1+L2 is \$429 [-\$5, \$931]; -\$2 [-\$5, \$0]; \$61 [-\$9, \$646] for the three generators. It is positive but not distinguishable from zero for the two Qwen generators and slightly negative for SmolLM2-360M, whose classifier leaves only

6 of its 195 hallucinations for the judge to catch, and the judge caught none. Under the fail-open policy, then, the judge does not demonstrably pay for itself for any generator.

The result depends on the timeout policy. If slow judge verdicts were awaited and credited, the stand-alone clearance of L3 would be 63 to 84% instead of 48 to 59%, the routed cascade would need $k^* = 15, 8, 21$, and the saving from adding the judge would be \$929 [\$274, \$1,861]; -\$2 [-\$5, \$0]; \$709 [\$130, \$1,575]. The judge's apparent value therefore depends on accepting latencies above 2 s. Residual hallucinations after L2 are few (19, 6, 26 of 300 questions), the 27-scenario grid varies cost parameters without reflecting sampling error, and the runs share their questions, so none of the differences between generators is a general finding. What the data do support is that the classifier is the dominant layer, the deterministic layer is nearly free but also nearly useless on this task, and the judge's marginal value is small and uncertain.

5.5 Evidence from public production traces

Request arrivals (BurstGPT).

The Azure OpenAI trace has 10.6 million requests over 231 recorded days, and 4.6% of requests failed (zero response tokens). The hour-of-day profile has a peak-to-trough ratio of 2.2, so the arrival process is clearly non-homogeneous, as the NHPP formulation assumes. Within an hour, counts are more variable than Poisson. The median variance-to-mean ratio of counts per bin (Poisson = 1) is 1.4 at 1 s, 5.3 at 10 s, 3.9 at 1 min and 5.1 at 10 min, so requests are clustered at every scale from 1 s upward, mildly at 1 s and strongly from 10 s. By Proposition 3, an unprotected hallucination stream with $p = 0.6$ would inherit about 3.5 of that dispersion at the 10-minute scale, whereas a cascade with a residual probability of 0.05 would retain about 1.2.

Day-to-day volume is extremely variable: the median day has 19,024 requests, with a range from 52 to 425,604. The median absolute day-to-day log change is 0.61, a factor of 1.8, and 73% of days differ from the previous day by more than 30%. Net of the local level (the day count over the mean of the 14 neighbouring days), the coefficient of variation is 1.22 in days 0 to

120 and 2.77 in days 225 to 334. The smallest coefficient of variation of any of the 163 14-day windows with all days above 1,000 requests is 0.44, against a median window value of 0.91. We therefore treat 0.44 as a floor for the variability of this trace, not an estimate comparable to the net-of-trend values of Section 5.6. The variability reflects traffic growth, campaigns, regime shifts and probably logging gaps as well as random fluctuation (the series also contains 7 near-duplicate day pairs, which we treat as a data artefact), so we use it as a stress test and not as an estimate of the dispersion of hallucination counts. In that stress test we replay the measured multipliers, rescaled to mean 1, on event streams with means of 10 and 120 per day. The Poisson plan is exceeded on 23% and 31% of days with the multipliers of days 0 to 120, and on 16% and 23% with those of days 225 to 334, which are inflated by regime shifts.

Service incidents (LLM providers).

Between March 2023 and August 2024 the three providers declared 457 incidents, or 0.83 per day. Incident starts are strongly non-homogeneous: the mean weekday count is 4.1 times the mean weekend count, and hour-of-day homogeneity is rejected for every provider and subset. A homogeneous Poisson model is rejected on every inter-arrival and dispersion test. Allowing for weekly and daily seasonality removes part of the lack of fit but not all of it. The daily index of dispersion is 1.53 and still 1.30 after stratifying by day of week ($p = 6 \times 10^{-6}$), and a negative-binomial fit to daily counts gives $r \approx 1.6$. Part of this over-dispersion probably reflects a changing incident rate over the 18 months, which a weekday-stratified test does not remove (Section 5.7 adjusts the baseline for it). The practical consequence at this low rate is small: the Poisson and negative-binomial 95% capacities coincide (3 incident per day), and the Poisson capacity is exceeded on 3.5% of days, below the 5% target. This agrees with Proposition 2, since $\phi = 1 + \lambda c^2$ remains modest, about 1.2 to 1.5 here, when λ is below one per day.

Incident duration, measured from declared start to resolution, serves as a proxy for human response time. It has a median of 1.0 h, a 90th percentile of 5.1 h and a 99th percentile of 34 h; the mean is 3.1 h and the coefficient of variation 3.5, so response times are heavy-tailed and far from exponential. The 95th percentile of the number of concurrently open incidents is 1,

the same as for a Poisson occupancy with that mean (1), and two or more incidents are open at once 0.7% of the time. A single standing responder therefore covers 95% of the time at provider level, and the heavy tail of durations accounts for the rare long overruns. These are provider-declared outages and degradations, not hallucinations, so they inform the arrival and service-time structure of an LLM incident stream but not its rate.

Dated hallucination cases (court decisions).

The only public, dated log of hallucination incidents we found is a curated database of court decisions in which a party relied on fabricated AI content (Charlotin 2026; 2,149 dated cases, 2023-04-14 to 2026-10-01). It is confined to the legal domain, the date is the decision date at day resolution, and detection and reporting bias is strong. Counts grew from a handful per month to over a hundred in 2025, so we analyse only the flatter period from October 2025 to August 2026 (1,527 cases, 6.3 per weekday), and exclude September and October 2026 because recent months are incomplete. In that period weekday counts are over-dispersed relative to Poisson (day-of-week-stratified index 1.40, $p = 1 \times 10^{-4}$; daily-rate coefficient of variation net of trend 0.24). Part of this comes from the 10 US federal-holiday weekdays; without them the index is 1.25 and the coefficient of variation 0.18, a weak over-dispersion. At about six cases per weekday the effect on capacity is small: the 95% capacity is 11 under Poisson and 12 under a negative binomial with the stratified index, and a Poisson plan against the local mean is exceeded on 6.3% of weekdays (5.6% with a causal plan). This is consistent with Proposition 2 and with the other low-rate streams. It is, to our knowledge, the only dispersion estimate for dated hallucination incidents that public data allow, subject to the caveats above.

5.6 Human triage workloads in public ticket data

The fourth family of public data concerns what happens after an incident has been detected: how tickets arrive at, and are worked by, human teams in real organisations. The streams are customer-support and IT-incident tickets and not LLM incidents, but they are the best public evidence we know of on the arrival variability and the service-time distribution of

human triage. Each stream is analysed for all work days and with holiday and low-volume days excluded by the rules of Appendix B; in Table 8 each cell reads "all days → excluded".

Table 8. Arrival structure of human-handled ticket streams (work days only; Poisson plan = 95% quantile of Poisson with the local mean)

Stream	Tickets (per work day)	Work-day-stratified dispersion index (includes slow trends)	Daily rate CV net of trend and weekday	Poisson plan exceeded: in-sample local mean	NB plan exceeded: in-sample	Poisson plan exceeded: causal	NB plan exceeded : causal
Software company (customer tickets 2017-23)	22,016 (13.1)	3.5 → 3.1	0.32 → 0.27	9.0% → 7.8%	3.4% → 4.0%	9.6% → 8.6%	1.9% → 2.5%
ServiceNow incidents (12-week segment, 2016)	22,035 (350.2)	25.2 → 5.9	0.25 → 0.10	35.0% → 20.0%	1.7% → 5.5%	15.0% → 6.7% (20 → 15 days)	0.0% → 0.0%
Italian software help desk (2010-12)	3,804 (5.3)	2.5 → 2.4	0.41 → 0.38	7.5% → 7.2%	3.7% → 3.8%	9.7% → 8.7%	2.9% → 2.8%

Each cell: all work days → low-volume or holiday days excluded (software company and ServiceNow: days below 35% of the rolling median, 62 and 5 days; Italian help desk: national holidays, 26 days). Target exceedance is 5%. The dispersion index includes slow trends, whereas the CV is net of a local level, so the two are not directly comparable. In-sample: the local mean is a two-sided leave-one-out mean over about 40 surrounding work days after removing the weekday effect, and c is estimated from the same days that are scored, so neither is a feasible forecast. Causal: trailing 20-work-day mean, c estimated on earlier days only, rolling origin (the weekday effects are estimated on the full sample). The software company works Sunday to Thursday; the other work weeks are Monday to Friday.

All streams are strongly non-homogeneous over the week and the day (weekend ticket rates are 0.09, 0.12 and 0.04 of work-day rates), and arrivals remain far from Poisson even after time-rescaling, because daily volumes vary beyond the weekly cycle. Net of slow trends and the weekday effect, the coefficient of variation of the daily rate is 0.32 → 0.27 for the software company, 0.41 → 0.38 for the Italian help desk and 0.25 → 0.10 for the ServiceNow segment, where holidays and bridge days account for most of the value. The dated hallucination cases of Section 5.5 give 0.24 → 0.18. These values are of the same order as our scenario values ($c = 0.3$ lies within the range after exclusions and $c = 0.44$ above it), though all of them come from non-LLM or legal-domain streams and we do not claim a universal value.

Because $\varphi = 1 + \lambda c^2$, the effect on capacity depends on the rate. Against the in-sample local mean, a Poisson-sized plan is exceeded on 7.8%, 7.2% and 20.0% of work days (about 14, 5 and 376 tickets per day) instead of 5%, and on 9.0%, 7.5% and 35.0% with all days. A causal plan, which is what an operator could actually use, is exceeded on 8.6% and 8.7% of days for the two long streams with the Poisson quantile, and on 2.5% and 2.8% with a negative-binomial quantile. The negative-binomial plan is thus somewhat conservative, for reasons we cannot identify from these data. The in-sample negative-binomial columns are a calibration check of Proposition 2 and not an out-of-sample test, since c is estimated on the days that are scored, and the causal evaluation of the 12-week ServiceNow segment covers only 15 days and is uninformative.

Service times are heavy-tailed (Appendix C, Table C1). Resolution time has a median of 168 h against a mean of 699 h for the software company, 22 h against 178 h for ServiceNow, and 94 h against 211 h for the Italian help desk. The coefficients of variation are 3.0, 3.0 and 1.3, where an exponential distribution has 1, and the Weibull shapes of 0.36 to 0.51 indicate a decreasing hazard: a ticket that has stayed open for a long time is likely to stay open much longer. Two consequences follow for capacity planning. Models that assume exponential service are not supported, which is one more reason to size capacity from the quantile of the arrival count and to treat the service-time tail explicitly and not through a mean-based utilisation. And in the one stream with handling-time data (the software company, where workload is the in-progress hours of the tickets created that day, attributed to the creation day) the 95th percentile of the daily workload is 3.2 times its mean, against 1.9 for the daily ticket count. Capacity measured in handling hours thus needs a larger multiple of the mean than capacity measured in arrivals, and the unit of triage capacity of Section 3.1 should be defined for a typical ticket with the workload tail added separately. In-progress time is calendar time in a workflow state and not effort, and the comparison is not detrended, so the ratio is indicative only.

For the software company and the Italian help desk, the time-average number of open tickets is 290 and 34, and the observed 95th percentile of open tickets is 20% and 16% higher

than that of a Poisson occupancy with the same mean. This describes a static backlog plan sized with the long-run mean and is not a test of Poisson arrivals, because both logs change in volume over their span and a slowly varying volume alone produces such an excess.

5.7 Typed failure models on public production data

We now fit the process classes of Section 3.6 to the two public failure datasets (the numbers are produced by `run_failure_models.py`).

Provider incidents: weekly seasonality, a drifting level and mild self-excitation.

The incident rate was not constant over the 18 months: monthly counts were 11 to 27 in the first twelve months and 27 to 54 afterwards (homogeneity across months rejected, $\chi^2 = 87$ on 17 df). With a fixed seasonal profile and a constant level, a self-exciting term improves the fit for every stream (Table C2; AIC gains of 9 to 26, branching ratio 0.11 to 0.21), but part of this is the slowly changing level, which the self-excitation term can mimic. Once the baseline is multiplied by a monthly level, the estimated branching ratio falls to 0.05 to 0.09 (mean cluster size 1.06 to 1.10 incidents, half-life 1.5 to 3.8 h), the AIC gains shrink to 2 to 4, and the optimistic time-rescaling test no longer rejects the NHPP for the combined stream ($p = 0.17$), for OpenAI, or for the major and critical subset. A weekday-and-hour profile with a drifting level therefore describes most of the data. A weak residual self-excitation remains, roughly one incident in 11 to 18 being followed by a clustered successor, with borderline evidence, and Anthropic's rescaling test still rejects the NHPP ($p = 0.016$). The implied long-window Fano factor of 1.15 (Proposition 6b) is below the stratified dispersion index of 1.30 reported in Section 5.5, so part of the measured over-dispersion is the drifting level. The conclusion for planning is modest: provider incidents need a seasonal, level-adjusted baseline, and whether they also cluster is weakly supported and, at this rate, of small consequence.

Serving failures: heavy-tailed episodes.

In the BurstGPT trace, failed requests are concentrated. Outage-like episodes, defined as minutes with more than 20% failed requests and at least five failures, merged when ten minutes

apart, cover 3.6% of active minutes and contain 68.1% of all 487,629 failures (26 to 96% across five definitions; Table C3). The 672 episodes have a median duration of 4 minutes (90th percentile 27, maximum 455) and heavy-tailed sizes: median 78 failed requests, mean 494, 99th percentile 10,419, maximum 27,908. A lognormal fit has $\sigma = 1.70$. The Hill estimator on the upper 20% gives a tail index of about 0.9, which at this sample size cannot distinguish a Pareto tail with infinite variance from a lognormal, so we do not read it as evidence of infinite variance. What matters for Proposition 6c is the second-moment ratio $E[X^2]/E[X]$ of episode sizes, which is 11,374 empirically and 5,838 under the lognormal fit. The Fano factor of failed-request counts that arrive in such episodes is thus in the thousands. Conditioning on daily request volume, which itself ranges from 52 to 425,604 requests, the Pearson dispersion of daily failed-request counts is 5,961, far from the value of about 1 for a Poisson-thinned stream. A Poisson description of failed requests is wrong by several orders of magnitude, even after allowing for traffic.

The episodes themselves are not Poisson either. They average 2.9 per day, with a daily dispersion index of 19.9 unconditionally and 16.3 given daily volume, and their inter-arrival times have a coefficient of variation of 4.2. A Hawkes process fits much better than an NHPP when the baseline is the hour-of-day profile (AIC 4143 $\rightarrow$ 2374; branching ratio 0.85, half-life 2.4 h), and also when the baseline is proportional to the hourly request volume (AIC 3366 $\rightarrow$ 2326; branching ratio 0.74), although the rescaled-gap test still rejects the Hawkes fit. The branching ratio depends on how episodes are delimited (Table C3): it stays between 0.52 and 0.85 across five definitions and the heavy tail of sizes persists, but the unconditional daily dispersion index of episode counts ranges from 2.1 to 19.9, and a long outage may be split into several fragments. Only the direction, clustering and heavy tails, is robust.

Capacity under the typed model.

To see what the typed description changes for a human-triage plan, we combine three human-handled classes for an illustrative organisation, using parameters from our estimates and treating the classes as independent. The first is hallucinations after the classifier-only cascade

$L1+L2$ ($\lambda_H = 12.7$ per day, from the Qwen2.5-0.5B run). Its daily-rate coefficient of variation c_H is assumed to lie between 0.3 and 0.45, close to the 0.27 and 0.38 estimated net of trend for the two long ticket streams (0.32 and 0.41 with all days), so $\phi_H = 1 + \lambda_H c_H^2$ is 2.14 to 3.57 by Proposition 2. The second is provider incidents affecting the organisation ($\lambda_P = 0.83$, $\phi_P = 1.30$ from the stratified index). The third is own serving-failure episodes (λ_S from the episode counts over the 231 recorded days), with $\phi_S = 2.1$ for the coarse definition, 19.9 for the fine one and 1.8 for the coarse definition conditioned on request volume.

Treating the total as one Poisson stream gives a 95% capacity of 23 units per day with the fine definition and 21 with the coarse one, at $c_H = 0.37$. The typed superposition requires 33 and 25, and the single-Poisson plan is exceeded on 15.4% and 12.6% of days instead of 5% (Monte Carlo of 400,000 days; effective dispersion $\phi_{\text{eff}} = 2.6$ to 5.7, Proposition 6a). Across the nine scenarios (three values of c_H and three treatments of the serving class), required capacity is 24 to 34 units against a single-Poisson plan of 21 to 23, which is 14 to 48% more, and the single-Poisson plan is exceeded on 10.4% to 17.2% of days. The hallucination class alone, whose dispersion is assumed, accounts for a gap of 21% (23 rather than 19 units). The provider class adds little, and the serving class widens the gap substantially only when it is treated as strongly over-dispersed (the fine episode definition, $\phi_S = 19.9$).

This scenario illustrates the size of the single-Poisson error under stated assumptions. It is not evidence for the typed model. The class means are not those of any real organisation, the comparator is a single Poisson (a negative-binomial plan on the total would close part of the gap), and a serving outage would plausibly raise hallucination and ticket volumes at the same time, which would widen it.

6. Discussion

6.1 Economic interpretation

The model can be read as a standard problem of production under uncertainty in which the firm's own technology generates the shocks. The standby human workforce is a safety stock

held against random errors, and its size is a quantile of the error count, fixed by the ratio of the price of capacity to the cost of a shortfall (Proposition 7). Automated screening is a capital-like input that lowers the mean and, through thinning, the dispersion of what the workforce must absorb (Proposition 3), so labour and screening are substitutes and the optimal mix responds to their relative price (Proposition 5). This gives the operational question a price-theoretic form: when labour becomes dearer, or inference cheaper, the firm should buy more screening, up to the point set by the floor on standing staff and by the indivisibility of capacity.

Two features distinguish the problem from textbook inventory theory. First, risk is correlated across the screening layers. The independence formula credits redundancy with a diversification benefit that positive association removes (Proposition 4), and the pricing exercise in Table E2 shows how costly that error is when the layers fail together. In our measurements the association was weak, but the share of hallucinations passing two layers still exceeded the independence prediction by a factor of 1.3 to 1.6 when both layers saw every answer and 1.9 to 2.3 when routing hid answers from the second layer. Second, the cost of automation is not only the compute bill. The verifier that reduced required capacity by 78 to 95% rejected 15 to 71% of correct answers, so part of the apparent saving is a transfer of cost from the firm's standby labour to customers and to the firm's own revenue. A planner who reports capacity savings without the false-rejection rate overstates the benefit, and the NLI classifier we tried first, which cut capacity by blocking nearly everything, is the limiting case.

The service level itself is an economic choice and not a technical constant. Proposition 7 shows that a target of 95% is optimal only if an unhandled incident costs twenty times the price of a capacity unit, and that moving to 99% requires a five-fold higher shortfall cost, \$15,000 instead of \$3,000 at the scenario wage. Firms that quote service levels without a shortfall cost are therefore leaving the economics implicit. The same proposition prices planning errors. Ignoring day-to-day variation in demand raised expected daily cost by 9.4% to 65.3% in our examples, and staffing at the mean load roughly doubled it, which shows that the second moment of demand is a first-order input to the staffing decision.

The extension of Section 3.8 shows that the assumption of full detection is not innocuous. When detection depends on reviewers whose attention is private, screening and vigilance are substitutes in effect: the layers that remove errors also make errors rarer among the items a person inspects, and a reviewer who is paid per error found has less reason to look hard. In the illustration of Table E4, a three-layer cascade removes about 95% of the surviving errors but the number of incidents handled falls by more than 99%, because attention falls from complete to about one in eight. The firm that watches handled incidents, or plans to maintain the status quo while the dashboard improves, mistakes lost detection for success. This is moral hazard in the textbook sense, with an unobservable action and an outcome that is only partly informative about it (Holmström 1979), and it matches the human-factors finding that reliance on automation erodes attention to the cases it passes (Parasuraman and Manzey 2010). Two remedies follow from Proposition 8: rewards or flat pay that do not depend on how often errors occur, and gates that raise the prevalence of errors in the cases people see. The second is the structure of the gate-and-review systems described in Section 2.6.

6.2 Implications for managers and regulators

For managers, the practical sequence follows from the model. Estimate the dispersion of incident counts before fixing a staffing plan, since a coefficient of variation of the daily rate of 0.3 to 0.44 was enough for a Poisson-sized plan to be exceeded on 23 to 28% to 30 to 32% of days without guardrails and 7 to 16% of days with the best cascade, against a 5% target. State the shortfall cost that the service-level target implies and check that it is plausible. Evaluate each guardrail layer by its marginal effect on required capacity net of its false rejections, latency and failures, and measure the dependence between layers before crediting a stack with its nominal effectiveness. In our data one verifier delivered most of the benefit, the deterministic layer was nearly inert, and the routed judge's saving was indistinguishable from zero (paired intervals for the daily saving \$429 [-\$5, \$931]; -\$2 [-\$5, \$0]; \$61 [-\$9, \$646]) at about three times the latency. Do not read a falling count of handled incidents as proof that screening works, because

part of the fall is lost detection (Table E4); audit a sample of unflagged items to measure what is missed. Where risk can be estimated before generation, as in the pre-generation scoring that J.P. Morgan researchers describe (Watson et al. 2024), send the riskiest cases to people, since a gate with $s^2 > \gamma$ raises both detection and the value of each reviewer's attention (Proposition 8).

Failure classes should be modelled separately and pooled at the ticket level: in our illustrative organisation a typed superposition required 14 to 48% more daily capacity than a single Poisson on the total, though this gap depends on assumed dispersion and is not a validated gain.

For regulators and auditors, the lesson concerns evidence on layered safeguards. Requirements that ask for several independent safeguards should ask for evidence of independence, since the product of stand-alone effectiveness is a lower bound on residual risk, not an estimate of it. Liability rules also enter the model directly, because they set the shortfall cost: a tenfold increase in the cost of an unhandled incident moves the optimal service level from 90% to 99% (Proposition 7), and the standby capacity and the screening depth that firms rationally hold move with it. Finally, public reporting of incident counts, with their dispersion and not only their totals, would let others test the assumptions that capacity planning rests on. We found that the public data for this are almost non-existent.

6.3 Limitations and future research

The economic content of the model is also limited. It is a partial-equilibrium, single-firm cost-minimisation problem and not a welfare analysis: the shortfall cost, the false-block cost and the residual-harm cost are scenario parameters, user harms and external effects are compressed into unit costs, and demand does not respond to quality or price. Reviewer behaviour is modelled in the simplest way (Section 3.8): a single attention choice with quadratic cost, a linear reward and no observability of missed errors, and the parameters of Table E4 are illustrative and not estimated. We have no data on reviewer attention, so the vigilance result is a theoretical prediction that remains to be tested. Other strategic actors are absent, so the model omits the

adaptation of adversaries to known screens. Outside Section 3.8, detection is assumed complete. These are natural extensions, and each would change the quantitative conclusions.

The most important limitation is that arrivals are simulated and that no operator log of hallucinations is public. Outcomes are measured on real model outputs, but arrival times are generated and each arrival replays a bootstrapped example, so the Poisson and time-rescaling checks of Section 5.3 verify the machinery and are partly true by construction. The public traces we use are real production data, but they record requests, service incidents and human tickets, not hallucinations. The one dated hallucination log we found, a database of court decisions, is domain-specific, has day resolution and carries strong detection and reporting bias. The burstiness of the replay is therefore a scenario informed by real streams and not estimated from them. Validation on proprietary incident logs, even a few months of daily counts of hallucination-related tickets, is the most valuable follow-up. A closely related assumption is that every hallucination passing the guardrails is detected and becomes one human-handled event. In practice only a fraction is noticed or reported, so the capacities we report are upper bounds that hold under full detection (Section 3.1), and the detection process deserves a model of its own.

The evaluation is small and partly post hoc. The three generators answered the same 300 questions, so their results are paired and not independent replications, and each layer estimate rests on 141 to 195 hallucinations and on a few to a few dozen residual ones. The extractive verifier was introduced after the NLI classifier proved uninformative, so the choice of L2 was made after seeing results on the same data. It is trained on SQuAD, which flatters it, and its threshold was not tuned. The judge is the same 1.5B model in every run, so one arm is self-judged. The analyses should be repeated on a held-out sample with a pre-registered layer configuration. The dependence estimates are imprecise: the association between L2 and L3 and the excess over the independence formula are weak, their bootstrap intervals include zero or one for two of the three generators, and the independence prediction is computed from the same sample's marginals. The judge results also assume that verdicts of calls slower than 2 s are discarded (Section 5.4), and the evaluation uses small open models, 0.4 to 1.5 billion parameters,

on one extractive task with CPU latencies. Absolute hallucination rates are far higher than for frontier models, and latencies on GPUs or hosted inference differ. The method transfers; the numbers do not. Ground truth rests on string and token-F1 matching against SQuAD gold answers, which can mark valid paraphrases as hallucinations and inflates the base rate, and the cost parameters and selection weights are assumptions, so only the labour-to-compute ratio and the sensitivity grid are informative.

The typed models and the public data carry their own caveats. The Hawkes and episode fits use a single kernel family, a fixed seasonal profile estimated from the same data and, for BurstGPT, an episode definition that matters quantitatively and may split a long outage into fragments; bootstrap p-values are bounded below by $1/(B + 1)$ and the time-rescaling p-values are optimistic. The capacity comparison of Section 5.7 combines classes from different organisations and datasets under independence, with an assumed hallucination dispersion and a single Poisson as comparator, and common-cause dependence between classes would widen the gap. The ticket streams are customer-support and IT incidents and not LLM incidents. Whether hallucination tickets share their arrival variability and service times is an assumption that is plausible for the coefficient of variation of the daily rate, which is invariant to independent thinning with constant p , and less so for service times, which depend on the verification task. Resolution times are calendar times that include waiting for customers or vendors and are not triage effort. Ticket volumes in two of the logs change over their span, the ServiceNow log supports only a 12-week segment, and the arrival-variability estimates rest on three streams. Isolated low-count days such as holidays and bridge days inflate dispersion that a local mean cannot absorb; we report results with and without them, using a volume rule for two streams and holiday calendars for two others, and the outcome-based volume rule is not applied to low-rate streams. The conservatism of the causal negative-binomial plan, which we cannot explain, shows that our variability estimates are not fully understood. BurstGPT has no calendar dates, a 104-day gap, near-duplicate day pairs and order-of-magnitude regime changes in daily volume, so its

day-level variability reflects growth and logging as well as chance, and we use it only as a stress test.

Future work could proceed along three lines. The first is validation on production incident logs from operators, which would allow the counting-process assumptions to be tested directly and the dispersion of hallucination counts to be estimated and not assumed. The second is to model detection and triage explicitly, including the delay between an incident and its handling, so that capacity can be expressed as a function of response-time targets and not only of daily counts. The third is to study how guardrail dependence changes with the generator, the domain and adversarial pressure, since the evidence here concerns one task and three small models.

7. Conclusion

Human backup for machine errors is a safety stock, and screening is the capital that economises on it. Under the stated assumptions the standby workforce should be a quantile of the daily incident count, with the service level implied by the cost of a shortfall; over-dispersion raises the required safety stock beyond what a Poisson model or the dispersion index suggests; the independence formula for stacked screens overstates diversification and understates residual risk; a single Poisson on all failures misdescribes every class; and with indivisible capacity and a floor on standing staff, the cost-minimising depth of screening is not fixed at three and responds to the relative prices of labour and compute. Because screening lowers the prevalence of errors that reviewers see, it also erodes their vigilance, so a falling count of handled incidents overstates success unless rewards or gates restore attention.

On the measured outputs of three small open models, a single task-specific verifier reduced required capacity by 78 to 95% while rejecting 15 to 71% of correct answers, and a routed judge did not demonstrably pay for itself under a 2 s fail-open timeout. Mild burstiness raised the exceedance rate of a Poisson-sized plan from 5% to roughly 23 to 32% without guardrails and 7 to 16% with the best cascade, and ignoring it raised expected cost by 9 to 65%

in our examples. Public production data corroborate the structure of the arrival processes, with clustered requests, seasonal incidents with a drifting level, and ticket streams whose net-of-trend daily-rate coefficient of variation is 0.10 to 0.38. Hallucination arrivals themselves are simulated and no operator log is public, so the key next step is to test these assumptions on production incident logs and to estimate the shortfall cost directly.

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Appendix A. Proofs

Proof of Proposition 1. (a) The $\text{Poisson}(\lambda)$ distribution is stochastically increasing in λ , so $P(X \leq k)$ is non-increasing in λ and the smallest k that reaches α cannot fall as λ rises. (b) $k^* = 0$ exactly when $P(X = 0) = e^{-\lambda} \geq \alpha$, that is, when $\lambda \leq -\ln \alpha$. (c) For λ in $[n, n + 1)$ with $n \geq 1$, $P(X > n)$ increases in λ , so its minimum on that interval is $P(\text{Poisson}(n) > n)$. The sequence $P(\text{Poisson}(n) > n)$ increases in n towards $1/2$, and its first term is $P(\text{Poisson}(1) > 1) = 1 - 2e^{-1} = 0.264$. At $\lambda = 3$ it equals $1 - P(X \leq 3) = 0.353$. (d) is the Cornish-Fisher expansion of the Poisson quantile. ■

Proof of Proposition 2. (a) is the law of total variance: $\text{Var}(X_d) = E[\lambda M] + \text{Var}(\lambda M) = \lambda + \lambda^2 c^2$. (b) For $\text{Poisson } Y_\lambda$, $\partial E[(Y_\lambda - k)^+] / \partial \lambda = P(Y_\lambda \geq k)$, which is non-decreasing in λ , so $\lambda \mapsto E[(Y_\lambda - k)^+]$ is convex. Jensen's inequality applied to the mixing variable gives $E[(X_d - k)^+] = E[g(\lambda M)] \geq g(\lambda)$, where $g(\lambda) = E[(Y_\lambda - k)^+]$. (c) If ϕ is bounded, the count is asymptotically normal with variance $\lambda\phi$. For fixed c , X_d/λ converges in distribution to M as λ grows, so the α -quantile of X_d is asymptotically $\lambda F_M^{-1}(\alpha)$, and the normal approximation fails by an amount linear in λ . ■

Proof of Proposition 3. Under binomial thinning with constant probability p , $\text{Var}(N_p) = p^2 \text{Var}(N) + p(1 - p)\mu$ by the law of total variance. Dividing by $E[N_p] = p\mu$ gives $pF + (1 - p) = 1 + p(F - 1)$. The statement about c follows because thinning multiplies the intensity of each day by p and leaves the multiplier M unchanged. We also verified the identity by simulation (checks/check_thinning.py). ■

Proof of Proposition 4. The bounds are the Fréchet-Bonferroni bounds for the probability that no layer fires: $1 - \sum c_i \leq P(\text{no layer fires}) \leq 1 - \max_i c_i$, with the lower bound truncated at zero. If the indicators of catching are positively associated, $P(\cap \{I_i = 0\}) \geq \prod P(I_i = 0)$ (Esary et al. 1967). Multiplying by λ_0 gives the stated bounds on λ_N , and Proposition 1(a) transfers them to k^* . ■

Proof of Proposition 5. (a) Table 3 is a counterexample to unimodality. (b) For $N > N_0$ the labour cost is constant at C_{hk_min} while compute cost increases in N , so such N are dominated

by N_0 . (c) Write $f(N) = g[N + \theta K(N)]$ with $K(N) = \max\{k_{\min}, k^*(\alpha, \lambda_N)\}$ non-increasing in N . The difference $f(N + 1) - f(N) = g[1 + \theta(K(N + 1) - K(N))]$ is non-increasing in θ , so f has decreasing differences in (N, θ) and the largest minimiser is non-decreasing in θ by Topkis' (1978) monotone comparative statics. (d) is immediate. ■

Proof of Proposition 6. (a) Variances of independent counts add, and the normal approximation of the quantile uses the total variance; the factor $\sqrt{\phi_{\text{eff}}}$ is the ratio of the resulting safety stock to that of a Poisson with the same mean. (b) The cluster generated by a background event is a Galton-Watson tree with $\text{Poisson}(\alpha)$ offspring. Its total progeny is $\text{Borel}(\alpha)$ with mean $1/(1 - \alpha)$ and second moment $1/(1 - \alpha)^3$, so $E[X^2]/E[X] = 1/(1 - \alpha)^2$, and the stationary variance-to-mean ratio of the count of a Hawkes process equals this value in the long-window limit. (c) is Wald's identity for the compound Poisson variance, $\text{Var}(N) = vE[X^2]$. ■

Proof of Proposition 7. (a) For integer k , $E[(X - k - 1)^+] - E[(X - k)^+] = -P(X > k)$, so $\Phi(k + 1) - \Phi(k) = C_h - C_u P(X > k)$. The right-hand side is non-decreasing in k , so Φ is convex, and the smallest minimiser is the smallest k with $P(X > k) \leq C_h/C_u$, that is, $P(X \leq k) \geq 1 - C_h/C_u$. Setting $\alpha = 1 - C_h/C_u$ gives $C_u = C_h/(1 - \alpha)$. (b) If $C_u \leq C_h$ then $\Phi(1) - \Phi(0) = C_h - C_u P(X > 0) \geq C_h - C_u \geq 0$, and convexity gives $k^\circ = 0$. (c) By Proposition 2(b) the stop-loss transform of the mixed-Poisson count is at least that of the Poisson count at every k , so a firm that uses the Poisson model underestimates the shortfall term of Φ at every capacity and, because the marginal benefit of capacity $C_u P(X > k)$ is perceived to be lower where the tail is lighter, tends to choose a smaller k ; Table E1 reports the resulting cost. (d) $\Phi^\circ(C_h) = \min_k [C_h k + C_u E(X - k)^+]$ is a minimum of functions linear in C_h , hence concave, and by the envelope theorem its derivative at a point where the minimiser is unique equals that minimiser. ■

Proof of Proposition 8. (a) On one item the reviewer maximises $r \cdot e \cdot 1\{\text{error}\} - \psi e^2/2$ in expectation, that is, $rpe - \psi e^2/2$ over $e \in [0, 1]$, which is strictly concave with maximiser $e = \min\{1, rp/\psi\}$. (b) Monotonicity in ρ is immediate, and $\rho = \lambda_N/M$ falls as layers are added. In the interior region $e\lambda_N = r\lambda_N^2/(\psi M)$. (c) $e = 1$ requires $rp \geq \psi$, and then the expected payment per day is $r \cdot \rho M \cdot 1 = (\psi/\rho)\rho M = \psi M$ at the smallest such reward, independent of ρ . (d) Without

the gate the expected number of errors found is $\lambda_N \cdot e$, with $e^* = rp/\psi$ in the interior. With the gate the reviewed set has γM items and $s\lambda_N$ errors, so prevalence is $\rho' = s\lambda_N/(\gamma M)$, attention is $e' = rp'/\psi$ and the expected number found is $s\lambda_N \cdot e' = rs^2\lambda_N^2/(\gamma M\psi)$. The ratio of the two is s^2/γ , which exceeds one if and only if $s^2 > \gamma$. ■

Appendix B. Public Data: Sources, Filters and Estimators

BurstGPT. The trace (Wang et al. 2025; CC BY 4.0) holds 10.6 million request records of regional Azure OpenAI ChatGPT and GPT-4 services at one-second resolution over 231 days, in two contiguous segments of 121 and 110 days separated by an unrecorded gap of 104 days. Timestamps are seconds from an undisclosed day 0, so weekday labels are unknown. Failed requests are rows with zero response tokens. We estimate the hour-of-day profile, the day-to-day variability of request volume (robustly, because volume grows by two orders of magnitude and contains regime shifts) and the within-hour Fano factor of counts at 1 s, 10 s, 1 min and 10 min. The stress test replays the measured daily multipliers, rescaled to mean 1, on event streams with a given mean.

Provider incidents. The dataset of LLM service outages and incident reports (Chu et al. 2025; CC BY 4.0) contains provider-declared incidents scraped from the status pages of OpenAI, Anthropic and Character.AI. We use the 457 incidents starting in March 2023, when all three pages are populated. We test Poisson counts per day and week (index of dispersion, also stratified by day of week), day-of-week and hour-of-day homogeneity, exponential inter-arrival times after time-rescaling with a fitted hour-of-week intensity, the distribution of incident durations as a proxy for human response time, and the number of concurrently open incidents against the $M/G/\infty$ prediction.

Ticket logs. The software-company data (Abdellatif 2026) contain 66,691 rows; we keep tickets of type "Ticket" created in 2017 or later and resolved as "Done", which leaves 22,016 tickets with creation, resolution and in-progress times. The work week is detected from the data as the five busiest days (Sunday to Thursday for this company, Monday to Friday for the others). The

ServiceNow log (Amaral et al. 2018; 24,918 incidents in 2016 to 2017) has 98.9% of its incidents opened between 29 February and 1 June 2016, so its arrival analysis uses only the middle weeks (7 March to 27 May 2016) and its concurrency is not analysed. The Italian help-desk log (Verenich 2016) contains 3,804 cases from 2010 to 2012. A High/Highest/Blocker subset of the software company is not analysed because its rate rises from about zero to eleven tickets per work day over 2017 to 2023, which reflects a change in the priority field.

Holiday and low-volume exclusions. Holidays and bridge days produce isolated low-count days that a local mean cannot absorb, so every arrival analysis is reported twice. For the software company and ServiceNow we exclude work days with fewer than 35% of the centred 41-work-day rolling median (62 and 5 days), a rule that is reliable only for high-rate streams. For the Italian help desk we exclude Italian national holidays and Easter Monday (26 days), and for the court cases US federal holidays (10 days).

Estimators. For each stream we estimate weekday-stratified over-dispersion, the daily-rate coefficient of variation net of slow trends (the Pearson dispersion of weekday counts against a leave-one-out local mean over about eight weeks, after removing the weekday effect), the exceedance rate of Poisson-sized capacity plans, the distribution of resolution and in-progress times (exponential, lognormal and Weibull fits compared by AIC), the daily workload safety factor (95th percentile over mean of handling hours, against arrivals) and the number of concurrently open tickets against the $M/G/\infty$ prediction. The Poisson plan is evaluated twice: against a two-sided local mean over about 40 surrounding work days (in-sample, not a feasible forecast) and causally, with the trailing 20-work-day mean and a coefficient of variation estimated on earlier days only (rolling origin). Weekday effects and the work-week definition are estimated on the full sample, so only the level and the coefficient of variation are strictly causal.

Court cases. The Charlotin database (Charlotin 2026; CC BY 4.0, accessed 6 October 2026) lists court decisions with AI-fabricated content. We use the decision date at day resolution for October 2025 to August 2026 and analyse weekday counts with the same stratified and local-

mean estimators as for tickets. September 2026 (95 cases) and one case in October 2026 are excluded because of reporting lag.

Typed failure models. The Hawkes estimator was checked by simulation (recovery of μ_0 , α and β from synthetic seasonal Hawkes data, and no false excitation when $\alpha = 0$; checks/check_hawkes_recovery.py). The parametric-bootstrap likelihood-ratio test uses 200 simulations under the NHPP, because the boundary $\alpha = 0$ invalidates the usual chi-square reference. For BurstGPT failures, outage-like episodes are minutes with more than 20% failed requests and at least five failures, merged when at most ten minutes apart. The Monte Carlo of Section 5.7 simulates 400,000 days (seed 0); the hallucination class uses the measured L1+L2 rate for Qwen2.5-0.5B and an assumed daily-rate coefficient of variation of 0.3 to 0.45, the provider class uses the rate and stratified dispersion index of Section 5.5, and the serving class uses the episode rate over the 231 recorded days with the dispersion index under three treatments.

Appendix C. Supplementary Tables

Table C1. Resolution and handling times of tickets (hours)

Stream / measure	n	Median	Mean	90th pct	99th pct	CV	Best fit (AIC) / Weibull shape
Software company: creation to resolution	22,016	167.8	698.8	1487	8806	3.0	weibull / 0.51
Software company: time in "in progress"	19,662	2.2	97.3	220	1532	3.6	weibull / 0.30
ServiceNow: opened to resolved	23,323	22.2	178.5	382	2849	3.0	weibull / 0.36
ServiceNow: Critical/High only	496	49.7	172.9	391	1086	2.0	weibull / 0.55
Italian help desk: first to last event	3,803	94.4	211.2	627	1008	1.3	weibull / 0.50

Calendar hours including waiting states (customer, vendor, queue); the in-progress measure counts only time in that workflow state. Exponential fits are rejected in every case ($KS\ p < 10^{-39}$).

Table C2. Seasonal NHPP versus seasonal Hawkes process for provider-declared incidents (March 2023 to August 2024)

Stream	Incidents	Baseline	Branching ratio α	Mean cluster size $1/(1-\alpha)$	Excitation half-life (h)	AIC gain over NHPP	LR test vs NHPP (bootstrap p)	Rescaled-gap KS p: NHPP $\rightarrow$ Hawkes
all providers	457	profile, constant level	0.16	1.19	7.1	19.7	23.7 (≤ 0.005)	$5 \times 10^{-4} \rightarrow 0.47$

Stream	Incidents	Baseline	Branching ratio α	Mean cluster size $1/(1-\alpha)$	Excitation half-life (h)	AIC gain over NHPP	LR test vs NHPP (bootstrap p)	Rescaled-gap KS p: NHPP $\rightarrow$ Hawkes
		profile $\times$ monthly level	0.07	1.07	2.9	3.5	7.5 (0.025)	0.166 $\rightarrow$ 0.91
OpenAI	280	profile, constant level	0.11	1.12	3.8	10.8	14.8 (≤ 0.005)	0.076 $\rightarrow$ 0.26
		profile $\times$ monthly level	0.05	1.06	1.5	2.6	6.6 (0.020)	0.135 $\rightarrow$ 0.82
Anthropic	141	profile, constant level	0.21	1.27	4.2	26.2	30.2 (≤ 0.005)	$1 \times 10^{-6} \rightarrow 0.13$
		profile $\times$ monthly level	0.09	1.10	3.8	2.2	6.2 (0.025)	0.016 $\rightarrow$ 0.19
major/critical (impact ≥ 2)	198	profile, constant level	0.13	1.14	5.8	9.4	13.4 (≤ 0.005)	$5 \times 10^{-4} \rightarrow 0.13$
		profile $\times$ monthly level	0.06	1.06	1.5	3.9	7.9 (0.015)	0.115 $\rightarrow$ 0.44

Baseline: weekday/weekend $\times$ hour-of-day profile estimated from the incident data and held fixed; the second row of each stream additionally multiplies it by a monthly level estimated from monthly counts. Exponential excitation kernel; bootstrap p from 200 parametric simulations under the corresponding NHPP (smallest possible value 0.005). KS: Kolmogorov-Smirnov test of time-rescaled gaps against $\text{Exp}(1)$ with the fitted compensator; the p-values are optimistic because the profile (48 slots), the monthly levels and the Hawkes parameters are all fitted on the same events.

Table C3. Sensitivity of the BurstGPT failure-episode model to the episode definition

Failure-share threshold	Min. failures per minute	Merge gap (min)	Episodes	Share of failures	Daily dispersion index	Hawkes α	Median / 99th pct size
20.0%	5	10	672	68.1%	19.9	0.85	78 / 10419
20.0%	5	60	211	88.9%	2.1	0.74	80 / 32372
10.0%	5	30	279	96.5%	3.2	0.66	61 / 30295
50.0%	5	10	192	25.6%	3.3	0.52	43 / 19484
20.0%	20	30	154	74.4%	5.2	0.78	259 / 32333

The first row (20%, 5, 10) is the definition used in the text. Hawkes fits use 30 bootstrap replications and hour-of-day baselines.

Appendix D. Reproducibility and Data Availability

All code, per-query model outputs, guardrail decisions and analysis scripts are provided in an anonymised repository, the link to which is supplied to the editors with the submission. The experiments run in the order collect (run_experiment.py collect), re-time with the extractive verifier (retime), analyse (run_experiment.py analyze, run_all_analyses.sh), summarise (summarize.py), compute the economic tables (econ_tables.py) and build the figures, tables and manuscript (make_figures.py, paper_tables.py, paper/econ/build.js). Every number in the text

and tables is generated from the resulting JSON files. The public datasets are not redistributed; they are available from their original sources under the licences stated in Section 4.5 and Appendix B. Generator and verifier checkpoints are the public Hugging Face releases named in Section 4. All simulations and bootstrap draws are seeded.

Appendix E. Use of Generative AI Tools

[Template to be completed by the authors in line with the journal's policy.] Generative AI tools were used in the development of the analysis code and in drafting and language editing of the manuscript. The authors specified the research design, reviewed and verified all code, analyses, numerical results and references, revised the text, and take full responsibility for the content of the paper. The tools are not listed as authors.